

Control vs Type 2 Diabetes Single-Cell Transcriptomics

Datasets, literature, learning roadmap and analysis plan

Compiled 2026-10-05. Every item carries a verification label: VERIFIED (read in paper text), ABSTRACT (abstract only), UNVERIFIED (from memory; check before use). GEO, NCBI and EBI pages could not be opened from the compiling environment, so no GEO sample metadata has been confirmed.

Contents: 1 GEO metadata verified (newest); 2 Better datasets; 3 More data sources; 4 Links; 5 How to download; 6 Dataset verification log; 7 Reading list part 1; 8 Reading list part 2 (gaps); 9 Mentor plan part 1; 10 Mentor plan part 2. Where documents conflict on datasets, the newest sections (1 to 3) take precedence.

GEO metadata read directly (2026-10-05): what is verified now, and what is still blocked

1. Download status

- www.ncbi.nlm.nih.gov is now reachable, so GEO series and sample records can be read. I read them (below).
- The data **files** are still not downloadable. GEO download links return HTTP 301 and redirect to <ftp.ncbi.nlm.nih.gov> (verified: `.../geo/download/?acc=GSE249089&format=file&file=...` redirects to `https://ftp.ncbi.nlm.nih.gov/geo/series/GSE249nnn/GSE249089/suppl/...`). That host returns 403 from this environment, so all 27 manifest files failed (0 bytes). figshare, Zenodo, CELLxGENE, EBI also still return 403.
- **To unblock:** add <ftp.ncbi.nlm.nih.gov> (and figshare.com, zenodo.org, cellxgene.cziscience.com, datasets.cellxgene.cziscience.com, www.ebi.ac.uk) to Allowed domains in the environment's Network access, then start a new session. Or run `scripts/download_manifest.py` on your own machine.

2. What I extracted instead (committed to `data/metadata/`)

File	Content
GSE221156_samples.csv	All 54 Bandesh GEO samples with disease state, sex, age, BMI, HbA1c, race, chemistry, islet center, cause of death
GSE268904_samples.csv	14 adipose snRNA-seq samples with sex, age, BMI, condition
GSE342773_samples.csv	39 samples: 20 bulk whole-tissue, 8 ex vivo explant, 11 snRNA 10x channels with donor assignment

3. Verified facts per dataset (from the GEO records)

Accession	What GEO says	Use
GSE221156 (Bandesh, PMID 41986506)	Islets from 17 T2D, 17 ND and 14 PD cadaveric donors, 10x scRNA-seq. Donor IDs listed by group	Main islet dataset
GSE342773	Visceral adipose from 21 Korean adults: Lean n=8 (HbA1c <6.5%, BMI <23), Obesity n=5, Obesity+T2D n=8. Bulk RNA-seq on 20 donors (8/5/7); snRNA-seq on 11 10x channels	Best adipose T2D case-control found: compare Ob vs Ob+T2D to control for BMI
GSE268904 (PMID 41040169)	14 deep subcutaneous adipose snRNA-seq samples, Göttingen: 4 lean, 7 obese, 3 obese diabetic	Too few T2D (n=3) for donor-level inference
GSE278526	scRNA-seq (not snRNA-seq) of adipose stromal cells, omental and subcutaneous, obese vs obese+T2D; 58 recruited, 10 profiled by scRNA-seq; 4 pooled libraries	Manifest wrongly calls it snRNA-seq; donor labels live in the barcode metadata file
GSE249089 (PMID 38297378)	84 METSIM participants, subcutaneous adipose snRNA-seq, 10x v3.1, 4 donors pooled per run and demultiplexed with Demuxlet; stated purpose includes obesity and T2D; raw files withheld for privacy	Large (n=84); T2D labels must come from the metadata file
GSE81608 (PMID 27667665)	Xin 2016, 1,492 cells, 1,600 samples; supplementary RPKM matrix	Accession now confirmed
GSE86473 (PMID 27864352)	A SuperSeries of Lawlor subseries (662 samples)	Confirmed as Lawlor
GSE154126 (PMID 32739450)	Avrahami: 1 newborn, 5 toddlers, 2 adolescents, 4 adult controls, 10 T2D; cells per donor listed (adult controls 22, 22, 37, 54; T2D 24, 13, 11, 30, 15, 23, 11, 19, 34, 24). Supplementary CPM and read-count matrices	Small matrices; very few cells per donor

4. Bandesh (GSE221156) design traps, from the sample table

- 54 libraries from 48 donors: 42 single-donor libraries and 12 libraries from 6 pooled donor pairs (Islet47/48, 57/58, 59/60, 70/71, 84/85, 118/119). Six donors (Islet70, 71, 84, 85, 118, 119) exist only in pooled libraries, so their cells need genotype or hashtag demultiplexing; check how the authors assigned them.

- Donors 47, 48, 57, 58, 59, 60 appear both as single libraries and in pools: treat them as one donor, not two.
- **Chemistry is confounded with disease.** Among single-donor libraries: T2D has 7 V2 and 7 V3; ND has 4 V2 and 13 V3; PD has 1 V2 and 10 V3. Include chemistry as a batch term or you will mistake it for disease.
- **Islet center varies:** Scharp-Lacy 22, Wisconsin 7, Southern California 5, Miami 3, UPenn 4 (3 of them T2D, none ND). Another batch factor.
- **Covariates (single-donor libraries, means):** ND age 45.4, BMI 28.4, HbA1c 5.2, 5/17 female. PD age 50.3, BMI 30.8, HbA1c 5.9, 3/11 female. T2D age 48.5, BMI 34.0, HbA1c 7.8, 6/14 female. BMI rises with disease state, so T2D effects partly overlap with obesity effects; model BMI and say so.
- One ND donor has no center listed (NA).
- The counts of 17/14/17 donors in the GEO series text are for donors; libraries differ, so always aggregate to donor level.

5. Corrections to earlier documents

- Xin accession GSE81608 is now verified (earlier marked unverified).
- Lawlor's GSE86473 is a SuperSeries; the cell data are in its subseries.
- Avrahami's cell counts per donor are now known: adult controls and T2D donors contribute only 11 to 54 cells each (the newborn contributes 84), so cell-type-specific pseudobulk will be very sparse.
- GSE278526 is scRNA-seq of stromal cells, not snRNA-seq as the manifest says.
- Bandesh details I earlier called "10x version not confirmed": both V2 and V3 chemistry are used.

6. Still unverified (hosts blocked)

E-MTAB-5061 (Segerstolpe), E-MTAB-15009 (muscle), HPAP/PANC-DB, PanKbase Zenodo record, CELLxGENE collections, HRA002549 and its figshare bundle.

Better public datasets for Control vs T2D (update, 2026-10-05)

This supersedes the dataset ranking in mentor_plan_part1.md and mentor_plan_part2.md. Labels: VERIFIED = read in paper/portal text; ABSTRACT = abstract only; UNVERIFIED = not checked. GEO, EBI and CELLxGENE pages themselves could not be opened from the compiling environment; accessions come from the papers' own data-availability statements.

What changed

Two larger, freely downloadable islet datasets turned up that beat the small plate-based sets for statistics, and one of them does not need an access application. The 2017-era datasets (Lawlor, Segerstolpe) drop to learning/replication only.

Ranked

Rank	Dataset	Tissue	Donors	Cells	Tech	Where	Verification
1	Bandesh et al. 2025/2026, Stitzel lab (Jackson Lab)	Human islets	48: 17 non-diabetic, 14 pre-diabetic, 17 T2D	245,878	droplet-based scRNA-seq (10x version not confirmed)	GEO GSE221156 ; BioProject PRJNA913127; CELLxGENE collection; code on Zenodo 14656366	VERIFIED (data-availability text)
2	PanKbase integrated map (Vu et al., bioRxiv 2026)	Human islets	140 donors, 191 samples: 69 controls, 12 autoantibody+, 11 pre-diabetes, 12 T1D, 36 T2D	448,935	integrated from HPAP, IIDP, Prodo and other sources	Final annotated object on Zenodo record 15596314; PanKbase portal	VERIFIED (paper text); preprint
3	HPAP / Elgamal 2023	Human islets	29 ND, 17 T2D (of 67)	192,203	HPAP scRNA-seq	PANC-DB; registration and data-use terms	VERIFIED
4	Skeletal muscle snRNA-seq, Hansen et al. 2025 (J Physiol)	Human skeletal muscle (non-islet)	controls n=10; T2D n=9	135,225 nuclei from 38 biopsies	snRNA-seq	ArrayExpress E-MTAB-15009	VERIFIED (data-availability text)
5	Lawlor 2017	Islets	5 ND, 3 T2D	1,050	Fluidigm C1	GEO GSE86473	VERIFIED
6	Segerstolpe 2016	Islets	6 healthy, 4 T2D	2,209	Smart-seq2	sandberg.cmb.ki.se/pancreas	VERIFIED counts
7	Avrahami 2020	Islets	10 adult T2D, 4 adult ND, 8 younger ND	not retrieved	Fluidigm C1	GEO GSE154126	VERIFIED donors

Leads, not verified: "Latent plasticity of the human pancreas" (57 donors across development, health, T2D; over 4 million cells and nuclei, doi:10.1016/j.cmet.2026.07.023, ABSTRACT); the sex-differences islet study (52 donors, scRNA+snATAC, ABSTRACT); the "largest single-cell islet dataset, 650,000 cells across 121 donors" alpha-to-beta transdifferentiation study (doi:10.1101/2025.02.14.638309, ABSTRACT); PBMC scRNA-seq in T2D (Front Endocrinol 2024, doi:10.3389/fendo.2024.1397661, ABSTRACT; blood, not a primary disease tissue).

What each is best for

- **Best for the main project: Bandesh GSE221156.** 17 vs 17 donors in one lab with a single processing pipeline, droplet data, GEO deposit, and a free CELLxGENE copy. The authors report 14 cell types present in every donor and 511 DEGs in T2D beta cells (VERIFIED, abstract/results text). The CELLxGENE collection is split into four instances (all cells, beta, alpha, delta), so you can start with a smaller beta-cell file.
- **Best for scale and validation: PanKbase.** 36 T2D and 69 control donors, but it is an integration of several sources including HPAP, so it is NOT independent of Elgamal/HPAP. Do not use HPAP and PanKbase as two separate replications.
- **Independence check:** donor overlap between Bandesh and PanKbase is unknown. Check donor IDs before claiming replication.
- **Best for learning:** Lawlor (small, plate-based, easy), then Bandesh beta-cell subset.
- **Best for advanced work:** PanKbase object (donor-level modelling across phenotypic groups); the muscle snRNA-seq set if you want a second tissue.

- **Adipose:** still no verified T2D-labelled human adipose set. A Broad Single Cell Portal adipose atlas exists (SCP1376, human and mouse white fat across body weight; obesity-focused, not T2D) and a human adipose SVF Drop-seq study (SCP133, 10 subcutaneous samples) exists, but neither is a T2D case-control design.

Cautions

- **Muscle study design:** 38 biopsies from 19 participants (10 control, 9 T2D) before/after 2 weeks of high-intensity training. That implies two biopsies per person: repeated measures. Use donor as a random effect or analyse the baseline biopsy only. Donor number is 19, not 38.
- **Pre-diabetes group (Bandesh):** you can drop it or model it as a third group; it is not "control".
- **Preprints:** Bandesh (bioRxiv, later EMBO J 2026) and PanKbase (bioRxiv) include different versions; use the published or latest version and note it.
- **Confounders:** age, BMI, sex, HbA1c, and isolation site differ by donor; check them in the supplementary donor tables before modelling.

How to download (untested)

- **Bandesh:** GEO page <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE221156> (supplementary files). Or open the CELLxGENE collection <https://cellxgene.cziscience.com/collections/58e85c2f-d52e-4c19-8393-b854b84d516e> and download each dataset as .h5ad (Download button on the dataset row). Python alternative: the cellxgene-census package (verify the current API in the official docs).
- **PanKbase:** <https://zenodo.org/records/15596314> (final annotated dataset); portal <https://pankbase.org>; analysis files <https://data.pankbase.org/analysis-sets/PKBDS1349YHGQ/>. Zenodo: wget the file links listed on the record page; check file sizes first.
- **Muscle:** <https://www.ebi.ac.uk/biostudies/ArrayExpress/studies/E-MTAB-15009> (Files tab).
- **Code for Bandesh pipeline:** <https://zenodo.org/records/14656366> (a ready reference pipeline to read alongside the paper).

Updated recommendation

1. Start on Lawlor only if you want a tiny warm-up; otherwise go straight to the Bandesh beta-cell subset (CELLxGENE, 17 vs 17 donors).
2. Main analysis: Bandesh, donor-level pseudobulk per cell type, T2D vs ND (and PD as a sensitivity group).
3. Replication: Elgamal/HPAP or the PanKbase object restricted to donors not in Bandesh; directional concordance of DEGs.
4. Secondary: Lawlor/Seegerstolpe/Avrahami only as exploratory support.
5. Verify GEO sample labels and donor tables before any inference.

Still open: a T2D-labelled adipose set, and 2022-2026 reviews on single-cell work in diabetes.

More data sources for Control vs T2D single-cell transcriptomics (2026-10-05)

Labels: VERIFIED = numbers read in paper text or abstract returned by the literature index; LEAD = title/abstract only, donor counts and accessions not checked; PORTAL = a place to search, listing seen only in search snippets. No GEO or portal page was opened (blocked). Where a manifest entry you uploaded (data/manifest.csv) overlaps, it is noted.

1. Network reality check (tested from the compiling session)

Every candidate host below returned HTTP 403 (egress policy) when tested: zenodo.org, cellxgene.cziscience.com, datasets.cellxgene.cziscience.com, cells.ucsc.edu, singlecell.broadinstitute.org, data.humancellatlas.org, pankbase.org, data.pankbase.org, hpap.pmacs.upenn.edu, www.ebi.ac.uk, www.ncbi.nlm.nih.gov, figshare.com. github.com answered but hosts no data here. So these sources are fine for you to download on your own machine, but the sandbox cannot fetch them until Network access is widened. Add all of the above to Allowed domains if you want me to download.

2. Islet datasets (human, T2D vs control) beyond the first list

Source	What it adds	Status
HCA Data Portal copies of islet studies: Segerstolpe (project ae71be1d-ddd8-4feb-9bed-24c3ddb6e1ad), Lawlor (c6ad8f9b-d26a-4811-b2ba-93d487978446), "Single cell RNA-seq of human pancreatic endocrine cells from juvenile, adult control and T2D donors" = Avrahami (99101928-d9b1-4aaf-b759-e97958ac7403), "Massively parallel single-cell RNA-seq of 26,677 pancreatic islet cells from healthy and T2D donors" (1c6a960d-52ac-44ea-b728-a59c7ab9dc8e)	A second, standardized place to get the same islet studies, plus one 26,677-cell healthy/T2D study I had not listed	LEAD: project IDs seen in the HCA portal update list; donor counts not checked. URL pattern: <a href="https://data.humancellatlas.org/explore/projects/<id>">https://data.humancellatlas.org/explore/projects/<id> ;
HPAP-derived papers listed on PANC-DB: Weng et al., Nat Genet 2023, "Integration of single-cell multiomic measurements across disease states with genetics identifies mechanisms of beta cell dysfunction in T2D" (PMC10550816); Diabetologia 2026, "Single-cell profiling of pancreatic islets maps subtype-associated molecular alterations in T2D" (PubMed 42678439); Sci Rep 2025, same-donor scRNA-seq and snRNA-seq (PubMed 41102292)	Methods papers and extra analyses on the same HPAP donors	LEAD; same donors as HPAP, so NOT independent of Elgamal/PanKbase
Human islet multiome / T2D genetics: Nat Genet 2021 chromatin accessibility (doi:10.1038/s41588-021-00823-0); bioRxiv single-cell multiome of islets and diabetes risk (PubMed 39149326)	scATAC plus RNA for the T2D GWAS link	LEAD
"Single-cell mRNA-regulation analysis reveals cell type-specific mechanisms of T2D" (Nat Commun 2025, doi:10.1038/s41467-025-65060-z)	Re-analysis of existing public islet datasets; useful as a cross-check of which public sets others combine	LEAD (abstract: "repurposed datasets"?). Check its methods for the list of GSE accessions
scRiskCell, iMeta 2025 (PMC12371254)	T2D islet risk-cell framework built on public data	LEAD
Imaging mass cytometry of human pancreas in T2D, Cell Rep 2021	Protein-level, not RNA; validation of composition changes	LEAD
"Single-cell atlas of human pancreatic islet and acinar endothelial cells in health and diabetes", Nat Commun 2025 (doi:10.1038/s41467-024-55415-3)	Endothelial cells in diabetes; 3 pancreases for enrichment	LEAD

3. Adipose tissue (the tissue your repo is named for)

Source	Details	Status
HRA002549 (GSA-Human, China)	Visceral adipose SVF scRNA-seq; paper (Chin Med J 2025, doi:10.3760/cma.j.cn112137-20250324-00714) compares macrophages in obesity and obesity with T2D. Your manifest also lists a processed figshare bundle	Abstract confirms the dataset is used for obesity vs obesity+T2D; group sizes not checked
METSIM snRNA-seq, GSE249089 (in your manifest)	Paper: Diabetol Metab Syndr 2025, doi:10.1186/s13098-025-01975-3, "snRNA-seq of subcutaneous adipose tissue from 84 individuals with MetS from the METSIM cohort"	VERIFIED from abstract; check whether T2D status is recorded per participant
Front Immunol 2026, "A mitochondrial-stress adipocyte-macrophage circuit sustaining metaflammation in human T2D adipose tissue" (doi:10.3389/fimmu.2026.1768845, PMC13181277)	Human T2D adipose; likely has a T2D vs control comparison	LEAD; open the data-availability section
Nat Metab 2026, "Defining the vascular niche of human adipose tissue across metabolic states" (doi:10.1038/s42255-026-01475-2)	Adipose endothelial cells across metabolic states	LEAD
HCA Data Portal adipose projects: "Single cell full-length transcriptome of human subcutaneous adipose tissue" (f77290ae-0d7b-4239-b0fe-3cf2c9e8858d); integrated single cell and spatial WAT map (57916660-af5a-44d5-a7a9-2e84b65f8a68)	Reference atlases, obesity/lean, not T2D case-control	LEAD
Single Cell Portal SCP1376 (human and mouse WAT atlas) and SCP133 (10 human subcutaneous SVF samples)	Reference; obesity-focused	PORTAL
Adipose snRNA-seq "Adipocyte subsets with distinct metabolic profiles" (bioRxiv 2025, doi:10.1101/2025.09.14.673351; 65,668 nuclei, SAT vs IAT)	Depot comparison	LEAD
Mouse db/db adipose snRNA-seq (bioRxiv 2024, doi:10.1101/2024.02.06.578860)	Mouse T2D model; hypothesis support only	Not human

4. Other metabolically relevant tissues

Tissue	Source	Status
Skeletal muscle	Hansen et al., J Physiol 2025: 10 controls, 9 T2D, 38 biopsies, 135,225 nuclei, ArrayExpress E-MTAB-15009 (repeated measures)	VERIFIED
Skeletal muscle	"Full-length single nuclei atlas of human skeletal muscle insulin resistance" (bioRxiv 2026, doi:10.64898/2026.02.26.708250; clamp-phenotyped)	LEAD
Kidney (diabetic kidney disease, T2D complication)	Nat Genet 2024, doi:10.1038/s41588-024-01802-x: 81 samples across healthy, diabetic and hypertensive kidneys (scRNA, snRNA, Visium, CosMx, snATAC)	VERIFIED from abstract
Kidney	snRNA-seq of 5 healthy vs 6 DKD kidneys, Diabetes 2025, doi:10.2337/db25-0272	VERIFIED from abstract; very small
Kidney	Early diabetic nephropathy scRNA landscape, listed on HCA (project 577c946d-6de5-4b55-a854-cd3fde40bff2); spatial DKD atlas (Nature 2026, doi:10.1038/s41586-026-10363-4); DKD snRNA+snATAC bioRxiv 2022 (doi:10.1101/2022.01.28.478204)	LEAD
Liver	Int J Biol Sci 2024, doi:10.7150/ijbs.99176, "Reconstruction of the hepatic microenvironment ... underlying type II diabetes through scRNA-seq"	LEAD; species and donor numbers unchecked
Liver (reference only)	Human liver sc/sn/spatial atlas (HepatoL Commun 2022, doi:10.1002/hep4.1854)	Not T2D-labelled

Tissue	Source	Status
Blood (PBMC)	Front Endocrinol 2024, doi:10.3389/fendo.2024.1397661; Front Immunol 2024, doi:10.3389/fimmu.2024.1501660	LEAD; blood is a poor proxy for islet/adipose biology, use for exploratory immune questions

5. Portals to search yourself (filter disease = type 2 diabetes mellitus)

- NCBI GEO DataSets: query like ("type 2 diabetes"[All Fields]) AND ("single cell" OR "single nucleus" OR "snRNA-seq") AND "Homo sapiens"[Organism]; read each Series Matrix file for diagnosis and donor columns.
- CZ CELLxGENE Discover (cellxgene.cziscience.com): filter Disease = type 2 diabetes mellitus, Organism = Homo sapiens. Downloads are .h5ad with standardized metadata. The Bandesh collection (58e85c2f-d52e-4c19-8393-b854b84d516e) is here.
- Human Cell Atlas Data Portal (data.humancellatlas.org): search "diabetes".
- Single Cell Portal (Broad): search "diabetes" or "adipose".
- PanKbase (pankbase.org) and PANC-DB (hpap.pmacs.upenn.edu): islet-focused, with donor metadata.
- Common Metabolic Diseases Genome Atlas, CMDGA (cmdga.org): appeared in search results; curated metabolic-disease genomics resource. I did not open it.
- ArrayExpress/BioStudies: search "type 2 diabetes single-cell".

6. Recent reviews on single-cell work in diabetes (the open item from before)

- "Single-Cell Multi-Omics in Type 2 Diabetes Mellitus: Revealing Cellular Heterogeneity and Mechanistic Insights", Int J Mol Sci 2025, doi:10.3390/ijms262211005 (PMC12652634). Review of single-cell multi-omics in islets. LEAD: read the abstract and skim the dataset tables.
- "Adipose tissue macrophage heterogeneity in the single-cell genomics era", Mol Cells 2024, doi:10.1016/j.mocell.2024.100031 (PMC10960114). Adipose macrophages; relevant to obesity and T2D.
- "Single cell and single nucleus RNA sequencing in liver tissues: applications and prospects", Front Genet 2026, doi:10.3389/fgene.2026.1781941. Liver methods, including model organisms.
- Best-practice reviews already in the reading list remain the core methods references.

7. What I recommend

1. Keep Bandesh (GSE221156) as the main dataset and PanKbase/HPAP as validation (see better_datasets.md).
2. For adipose, rank by credibility: first METSIM GSE249089 (84 participants; check T2D labels), then HRA002549 (obesity vs obesity + T2D), then the 2026 Front Immunol human T2D adipose paper once you read its data-availability section.
3. For a non-islet T2D tissue with verified numbers, use the skeletal muscle snRNA-seq (E-MTAB-15009), handling repeated measures.
4. Treat kidney, liver and PBMC as exploratory.
5. Before using any source, open its GEO/portal page and confirm diagnosis labels and donor counts. I could not.

Links: datasets and priority papers

Links built from accessions/DOIs/PMIDs returned by searches or read in paper text. I could not open GEO, ArrayExpress or PANC-DB from my sandbox, so the dataset pages themselves are unchecked; click through and confirm sample labels.

Datasets

Dataset	Accession	Link	Status
Lawlor 2017 (5 ND, 3 T2D)	GEO GSE86473	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE86473	Accession read in paper
Lawlor 2017 raw	SRA SRP075970, BioProject PRJNA323853	https://www.ncbi.nlm.nih.gov/bioproject/PRJNA323853	Accession read in paper
Avrahami 2020 (10 T2D adults)	GEO GSE154126	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE154126	Accession read in paper
Segerstolpe 2016 (6 healthy, 4 T2D)	Web portal	http://sandberg.cmb.ki.se/pancreas	URL read in paper
Segerstolpe 2016	ArrayExpress E-MTAB-5061	https://www.ebi.ac.uk/biostudies/arrayexpress/studies/E-MTAB-5061	UNVERIFIED accession (from memory)
Xin 2016	GEO GSE81608	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE81608	UNVERIFIED accession (from memory)
HPAP (Elgamal 2023)	PANC-DB	https://hpap.pmacs.upenn.edu	UNVERIFIED URL (from memory); paper says raw FASTQ come from the PANC-DB portal
Bandesh 2025/2026	see paper's data availability	https://pmc.ncbi.nlm.nih.gov/articles/PMC11785113/	Accession not retrieved

Papers (reading order)

#	Paper	DOI link	PMC (free full text)
1	Luecken & Theis, Current best practices in scRNA-seq analysis (Mol Syst Biol 2019)	https://doi.org/10.15252/msb.20188746	https://pmc.ncbi.nlm.nih.gov/articles/PMC6582955/
2	Best practices for single-cell analysis across modalities (Nat Rev Genet 2023)	https://doi.org/10.1038/s41576-023-00586-w	https://pmc.ncbi.nlm.nih.gov/articles/PMC10066026/
3	Confronting false discoveries in single-cell DE (Nat Commun 2021)	https://doi.org/10.1038/s41467-021-25960-2	https://pmc.ncbi.nlm.nih.gov/articles/PMC8479118/
4	Benchmarking differential state methods, multi-subject scRNA-seq (Brief Bioinform 2022)	https://doi.org/10.1093/bib/bbac286	https://pmc.ncbi.nlm.nih.gov/articles/PMC9487674/
5	Design and power analysis for multi-sample single cell genomics (Nat Commun 2021)	https://doi.org/10.1038/s41467-021-26779-7	https://pmc.ncbi.nlm.nih.gov/articles/PMC8595682/
6	Elgamal et al., Integrated map of islet gene expression (Diabetes 2023)	https://doi.org/10.2337/db23-0130	https://pmc.ncbi.nlm.nih.gov/articles/PMC10588282/
7	Lawlor et al. (Genome Res 2017)	https://doi.org/10.1101/gr.212720.116	https://pmc.ncbi.nlm.nih.gov/articles/PMC5287227/
8	Segerstolpe et al. (Cell Metab 2016)	https://doi.org/10.1016/j.cmet.2016.08.020	https://pmc.ncbi.nlm.nih.gov/articles/PMC5069352/
9a	Benchmarking atlas-level data integration (Nat Methods 2022)	https://doi.org/10.1038/s41592-021-01336-8	https://pmc.ncbi.nlm.nih.gov/articles/PMC8748196/
9b	Batch correction methods are often poorly calibrated (Genome Res 2025)	https://doi.org/10.1101/gr.279886.124	https://pmc.ncbi.nlm.nih.gov/articles/PMC12315870/
10a	Milo (Nat Biotechnol 2022)	https://doi.org/10.1038/s41587-021-01033-z	https://pmc.ncbi.nlm.nih.gov/articles/PMC7617075/
10b	scCODA (Nat Commun 2021)	https://doi.org/10.1038/s41467-021-27150-6	https://pmc.ncbi.nlm.nih.gov/articles/PMC8616929/

Extra, as you reach those weeks

Paper	DOI link	PMC
Avrahami et al. 2020	https://doi.org/10.1016/j.molmet.2020.101057	https://pmc.ncbi.nlm.nih.gov/articles/PMC7471622/
Xin et al. 2016	https://doi.org/10.1016/j.cmet.2016.08.018	none returned

Paper	DOI link	PMC
Bandesh et al. (bioRxiv 2025)	https://doi.org/10.1101/2025.01.17.633590	https://pmc.ncbi.nlm.nih.gov/articles/PMC11785113/
Bandesh et al. (EMBO J 2026)	https://doi.org/10.1038/s44318-026-00744-w	https://pmc.ncbi.nlm.nih.gov/articles/PMC13226668/
Islet cell identity genesets (Nat Commun 2022)	https://doi.org/10.1038/s41467-022-29588-8	https://pmc.ncbi.nlm.nih.gov/articles/PMC9019032/
decoupleR (Bioinform Adv 2022)	https://doi.org/10.1093/bioadv/vbac016	https://pmc.ncbi.nlm.nih.gov/articles/PMC9710656/
Integrated single cell and spatial map of human white adipose tissue (Nat Commun 2023)	https://doi.org/10.1038/s41467-023-36983-2	https://pmc.ncbi.nlm.nih.gov/articles/PMC10017705/

How to download the datasets

These commands are untested: GEO, EBI and SRA were blocked from the environment where this was compiled. Accessions are the ones read in the papers (GSE86473, GSE154126, SRP075970). The FTP path pattern and tool names are from memory. If a path fails, copy the exact link from the GEO page.

Start here: Lawlor 2017, processed (GEO GSE86473)

Option A: browser (simplest)

Open <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE86473>. Scroll to **Supplementary file** at the bottom and use the (ftp) or (http) links. Also open **Series Matrix File(s)**: it carries the sample labels (control vs T2D, donor ID), which you must confirm.

Option B: command line

GEO series folders follow a fixed pattern: the last three digits of the number become nnn.

```
mkdir -p data/GSE86473 && cd data/GSE86473
wget -r -np -nH --cut-dirs=4 -R "index.html*" \
  https://ftp.ncbi.nlm.nih.gov/geo/series/GSE86nnn/GSE86473/suppl/
wget https://ftp.ncbi.nlm.nih.gov/geo/series/GSE86nnn/GSE86473/matrix/
```

Option C: Python

```
import GEOParse # pip install GEOParse
gse = GEOParse.get_GEO("GSE86473", destdir="data/")
meta = gse.phenotype_data # sample table: look for diagnosis and donor columns
gse.download_supplementary_files(directory="data/GSE86473_suppl")
```

Option D: R

GEOquery::getGEO("GSE86473") and getGEOSupFiles("GSE86473").

Most supplementary files of this kind are plain-text count tables. Load with `pandas.read_csv(..., sep="\t", index_col=0)`, then build an AnnData object with `anndata.AnnData(df.T)`. Check first whether rows are genes or cells.

Raw reads (FASTQ): only if you need to realign

Raw data for Lawlor is in SRA under SRP075970 (BioProject PRJNA323853).

```
# install sra-tools: conda install -c bioconda sra-tools
# get the run list: SRA Run Selector -> search SRP075970 -> download Accession List
prefetch --option-file SRR_Acc_List.txt
fasterq-dump --split-files SRRxxxxxxx
```

Raw FASTQ files are large: check disk space first. For a first project, use the processed matrix.

Avrahami 2020 (GEO GSE154126)

Same steps as above. The FTP folder is `.../geo/series/GSE154nnn/GSE154126/`.

Segerstolpe 2016

- Easiest: the authors' portal at <http://sandberg.cmb.ki.se/pancreas>.
- ArrayExpress E-MTAB-5061 (accession unverified): open the study page, go to the Files tab, download the processed counts file and the metadata (SDRF) file. The SDRF is the sample table: look in it for the disease label and the donor.

HPAP (main project dataset)

HPAP is not a one-click download.

1. Go to PANC-DB (<https://hpap.pmacs.upenn.edu>, URL from memory) and register.
2. Accept the data-use terms. This may need an institutional signature or PI approval.
3. Download donor metadata and the scRNA-seq files. The Elgamal paper says raw FASTQ files come from there.
4. Ask your PI now: approval can take time.

Processing 60-plus donors from FASTQ also needs a cluster. Check whether Khalifa University HPC is available to you.

Before you analyze

- Metadata first: build a donor table (donor ID, diagnosis, sex, age, BMI, batch). Count donors per group, not cells.
- Record every file's source URL and download date in `data/README.md`.
- Keep raw downloads read-only and never edit them.

Control vs T2D single-cell datasets: verification log (work in progress)

Status as of 2026-10-05. Only facts read from a primary source (paper full text on PMC) are marked VERIFIED. GEO, NCBI, and EBI were unreachable from the sandbox, so GEO pages were NOT read directly. Anything marked UNVERIFIED comes from memory or abstracts and must be checked on GEO/ArrayExpress before use.

Pancreatic islets (human, scRNA-seq)

Dataset	Paper	Donors	Cells	Tech	Accession	Status
Lawlor 2017	Genome Res, doi:10.1101/gr.212720.116	8 (5 ND, 3 T2D)	1,050 (622 ND, 428 T2D)	Fluidigm C1, ~3M reads/cell	Raw: SRA SRP075970 / PRJNA323853. Processed: GEO GSE86473	VERIFIED (paper text)
Segerstolpe 2016	Cell Metab, doi:10.1016/j.cmet.2016.08.020	10 (6 healthy, 4 T2D)	2,209	Smart-seq2, ~750k reads/cell	Portal: sandberg.cmb.ki.se/pancreas. ArrayExpress E-MTAB-5061 UNVERIFIED	Counts VERIFIED; accession UNVERIFIED
Avrahami 2020	Mol Metab, doi:10.1016/j.molmet.2020.101057	22 (1 newborn, 5 toddler, 2 adolescent, 4 ND adult, 10 T2D adult)	not retrieved	Fluidigm C1, SMART-seq, median 1.7M aligned reads	GEO GSE154126	VERIFIED except cell count
Elgamal 2023 (HPAP integrated map)	Diabetes, doi:10.2337/db23-0130	67 downloaded (29 ND, 17 T2D, 10 T1D, 9 ND Aab+); 65 in final map	192,203	HPAP scRNA-seq; raw FASTQ from PANC-DB	PANC-DB portal (data access terms apply)	VERIFIED (paper text); 10x vs other chemistry not confirmed
Xin 2016	Cell Metab, doi:10.1016/j.cmet.2016.08.018	ND and T2D organ donors (counts not retrieved)	1,492 (alpha, beta, delta, PP)	not confirmed	GEO GSE81608 UNVERIFIED	Abstract only
Bandesh 2025/2026	bioRxiv doi:10.1101/2025.01.17.633590; EMBO J doi:10.1038/s44318-026-00744-w	not retrieved	not retrieved	not retrieved	not retrieved	Title/abstract only; read the paper

Notes for a beginner:

- Lawlor, Segerstolpe, Xin are small (4-10 donors, plate-based). Good for learning and for quick replication, weak for donor-level statistics.
- Elgamal/HPAP has the most T2D donors (17 vs 29 ND) and is the best candidate for the main analysis, but needs PANC-DB access and re-processing from FASTQ.

Adipose tissue

No human adipose scRNA/snRNA dataset with explicit T2D vs control labels has been verified yet. Leads (all UNVERIFIED for T2D labels, donor counts, and accessions):

- Single-cell analysis of human adipose tissue, depot and disease specific cell types, Nat Metab 2020, doi:10.1038/s42255-019-0152-6.
- Human adipose snRNA-seq study using dataset HRA002549 (GSA, China), obesity vs obesity+T2D macrophage analysis, doi:10.3760/cma.j.cn112137-20250324-00714.
- Insulin-resistance snRNA-seq in subcutaneous adipose, PLoS Genet 2020, doi:10.1371/journal.pgen.1009018.
- Emont et al. 2022 (Nature) human/mouse adipose snRNA-seq: obesity-focused, not T2D.

Benchmarks / methods leads (for part 2, not yet read)

- Benchmarking differential state methods for multi-subject scRNA-seq: Nat Commun/Brief Bioinform, doi:10.1093/bib/bbac286 (pseudobulk performed best).
- Pseudobulk with offsets vs GLMM: Bioinformatics, doi:10.1093/bioinformatics/btae498.

Next steps

1. Confirm GSE/E-MTAB accessions and per-sample labels on GEO/ArrayExpress (needs network access to those hosts).
2. Read Bandesh 2025/2026 and Elgamal methods for chemistry, QC, and donor metadata.
3. Search Single Cell Portal, HCA, and CELLxGENE for T2D-labelled adipose/liver/muscle data.

Reading list: Control vs T2D single-cell transcriptomics

Every entry below was returned by a literature index search (title, DOI, abstract) on 2026-10-05. I did NOT read full texts except where noted in datasets_verified_so_far.md. Takeaways are quoted or paraphrased from abstracts only. Verify details before citing. "Order" is a suggested reading order.

A. Foundational: workflow, QC, normalization

Order	Paper	DOI	Why read it
1	Current best practices in single-cell RNA-seq analysis: a tutorial (Mol Syst Biol 2019)	10.15252/msb.20188746	Start here. End-to-end workflow and the decisions at each step.
2	Tutorial: guidelines for the computational analysis of single-cell RNA sequencing data (Nat Protoc 2021)	10.1038/s41596-020-00409-w	Second pass with a protocol focus.
3	Bioconductor workflow for single-cell RNA sequencing: normalization, dimensionality reduction, clustering, lineage inference (F1000Res 2017)	10.12688/f1000research.12122.1	R/Bioconductor route if you choose R over Python.
4	Analysis and visualization of single-cell sequencing data with Scanpy and MetaCell: a tutorial (Methods Mol Biol 2024)	10.1007/978-1-0716-3642-8_17	Python/Scanpy route, matches your background.

B. Quality control: ambient RNA and doublets

Paper	DOI	Why read it
Decontamination of ambient RNA in single-cell RNA-seq with DecontX (Genome Biol 2020)	10.1186/s13059-020-1950-6	Ambient RNA method.
Understanding and mitigating the impact of ambient mRNA contamination (PLoS One 2025)	10.1371/journal.pone.0332440	Abstract states ambient transcripts appear among DEGs; compares CellBender and SoupX. Directly relevant to islets (insulin/glucagon spillover).
FastCAR: fast correction for ambient RNA to facilitate differential expression (BMC Genomics 2023)	10.1186/s12864-023-09822-3	Abstract: ambient RNA levels are highly sample-specific, which matters for case vs control.
Souporcell: clustering by genotype without reference genotypes (Nat Methods 2020)	10.1038/s41592-020-0820-1	Donor demultiplexing and cross-genotype doublets.

C. Integration and batch correction (needed because donors are batches)

Paper	DOI	Takeaway from abstract
A benchmark of batch-effect correction methods for scRNA-seq (Genome Biol 2020)	10.1186/s13059-019-1850-9	Recommends Harmony, LIGER, Seurat 3.
Benchmarking atlas-level data integration in single-cell genomics (Nat Methods 2022)	10.1038/s41592-021-01336-8	BBKNN, Scanorama, scVI do well on complex tasks.
Batch correction methods ... are often poorly calibrated (Genome Res 2025)	10.1101/gr.279886.124	Recommends only Harmony. Read next to the two above; they disagree.
Probabilistic harmonization and annotation with deep generative models, scVI/scANVI (Mol Syst Biol 2021)	10.15252/msb.20209620	scVI/scANVI model reference.

D. Case-control statistics (the most important section for your design)

Paper	DOI	Takeaway from abstract
Confronting false discoveries in single-cell differential expression (Nat Commun 2021)	10.1038/s41467-021-25960-2	Why cell-level tests inflate false positives. Read first.
Differential gene expression analysis for multi-subject scRNA-seq with aggregateBioVar (Bioinformatics 2021)	10.1093/bioinformatics/btab337	Naive testing gives many false discoveries; pseudobulk has better FDR control.
A balanced measure shows superior performance of pseudobulk methods over mixed models and pseudoreplication (Nat Commun 2022)	10.1038/s41467-022-35519-4	Supports pseudobulk.

Paper	DOI	Takeaway from abstract
Benchmarking methods for detecting differential states between conditions from multi-subject scRNA-seq (Brief Bioinform 2022)	10.1093/bib/bbac286	Abstract: pseudo-bulk methods performed generally best.
Pseudobulk with proper offsets has the same statistical properties as GLMMs (Bioinformatics 2024)	10.1093/bioinformatics/btae498	Counterpoint on mixed models.
Valid post-clustering differential analysis for scRNA-seq (Cell Syst 2019)	10.1016/j.cels.2019.07.012	Double-dipping after clustering.

E. T2D single-cell papers (islets)

Paper	DOI	Role
Xin et al., RNA sequencing of single human islet cells reveals T2D genes (Cell Metab 2016)	10.1016/j.cmet.2016.08.018	1,492 cells, ND vs T2D donors (abstract).
Segerstolpe et al., Single-cell transcriptome profiling of human islets in health and T2D (Cell Metab 2016)	10.1016/j.cmet.2016.08.020	6 healthy, 4 T2D, 2,209 cells (read in full).
Lawlor et al., Single-cell transcriptomes identify human islet cell signatures ... in T2D (Genome Res 2017)	10.1101/gr.212720.116	5 ND, 3 T2D, 1,050 cells (read in full).
Avrahami et al., Single-cell transcriptomics of human islet ontogeny ... beta-cell dedifferentiation in T2D (Mol Metab 2020)	10.1016/j.molmet.2020.101057	22 donors incl. 10 T2D (read in full).
Camunas-Soler et al., Pancreas patch-seq links physiologic dysfunction in diabetes to transcriptomic phenotypes (Cell Metab 2020)	10.1016/j.cmet.2020.04.005	Links function to expression; advanced.
Elgamal et al., An integrated map of cell type-specific gene expression in pancreatic islets (Diabetes 2023)	10.2337/db23-0130	192,203 cells, 17 T2D vs 29 ND (read in full). Best model for your own pipeline.
Bandesh et al., Single-cell decoding of human islet cell type-specific alterations in T2D (bioRxiv 2025; EMBO J 2026)	10.1101/2025.01.17.633590 ; 10.1038/s44318-026-00744-w	Recent, large; read methods.
Single-cell mRNA-regulation analysis reveals cell type-specific mechanisms of T2D (Nat Commun 2025)	10.1038/s41467-025-65060-z	Re-analysis of public islet scRNA-seq.
Human pancreatic alpha-cell heterogeneity ... SMOC1 as a beta-cell dedifferentiation gene (Nat Commun 2025)	10.1038/s41467-025-62670-5	Combines scRNA-seq and snRNA-seq.
Meta-analysis of single-cell expression of human alpha and beta cells in T2D (J Diabetes 2021)	10.1111/1753-0407.13236	Example of combining the small public datasets.

F. Adipose tissue (relevant to this repo)

Paper	DOI	Note
Single-cell analysis of human adipose tissue identifies depot and disease specific cell types (Nat Metab 2020)	10.1038/s42255-019-0152-6	scRNA-seq of SVF from obese individuals.
An integrated single cell and spatial transcriptomic map of human white adipose tissue (Nat Commun 2023)	10.1038/s41467-023-36983-2	Integrates ten studies; >60 subpopulations; a reference for annotation.
Human subcutaneous and visceral adipocyte atlases (Nat Genet 2025)	10.1038/s41588-024-02048-3	snRNA-seq depot atlases.
Spatial mapping reveals human adipocyte subpopulations with distinct sensitivities to insulin (Cell Metab 2021)	10.1016/j.cmet.2021.07.018	Insulin sensitivity link.
Integrative single-cell analysis of metabolic syndrome in adipose tissue (Diabetol Metab Syndr 2025)	10.1186/s13098-025-01975-3	Metabolic-syndrome snRNA-seq reanalysis.
Robust snRNA-seq reveals depot-specific dynamics in adipose remodeling during obesity (eLife 2025)	10.7554/eLife.97981	Mouse-focused nuclei isolation method; check species before use.
Causal effect of obesity on prediabetes and insulin resistance (PLoS Genet 2020)	10.1371/journal.pgen.1009018	Bulk plus snRNA-seq, insulin resistance.

G. Atlases and portals

Paper	DOI
The Tabula Sapiens (Science 2022)	10.1126/science.abl4896
Tabula Sapiens 2.0 (bioRxiv 2024)	10.1101/2024.12.03.626516

Gaps (not yet found or not yet read)

- Experimental design and power for scRNA-seq case-control; sequencing technology comparisons (Smart-seq2 vs 10x vs snRNA-seq).
- Cell type annotation tools; cell-cell communication; pathway/GSEA on pseudobulk; trajectory inference.
- Your remaining items 3 to 5 (message was cut off).

Reading list, part 2: filling the gaps

Companion to literature_reading_list.md. Same caveat: every entry was returned by a literature-index search (title, DOI, abstract) on 2026-10-05; I did not read full texts. Takeaways come from abstracts only.

H. Experimental design and power

Paper	DOI	Why read it
Design and power analysis for multi-sample single cell genomics experiments (Nat Commun 2021)	10.1038/s41467-021-26779-7	Abstract: models sample size, cells per individual and depth. Core design paper.
Hierarchicell: power for differential expression with single-cell data (BMC Genomics 2021)	10.1186/s12864-021-07635-w	Abstract: existing calculators focus on total cells, not independent units (donors), and overestimate power.
Experimental design for single-cell RNA sequencing (Brief Funct Genomics 2018)	10.1093/bfpg/elx035	General design considerations.
Quantifying the tradeoff between sequencing depth and cell number in scRNA-seq (bioRxiv)	10.1101/762773	Budget allocation: cells vs depth.
Optimal design of scRNA-seq experiments for cell-type-specific eQTL analysis (Nat Commun 2020)	10.1038/s41467-020-19365-w	Abstract: different sample/cell/read designs can give similar power.
Robustness of scRNA-seq for identifying differentially expressed genes (BMC Genomics 2023)	10.1186/s12864-023-09487-y	Small cell numbers per cluster.

I. Technology comparison (matters for pooling Smart-seq2, C1, 10x, snRNA-seq datasets)

Paper	DOI	Takeaway from abstract
Direct comparative analysis of 10X Genomics Chromium and Smart-seq2 (Genomics Proteomics Bioinformatics 2021)	10.1016/j.gpb.2020.02.005	Smart-seq2 detected more genes.
Comparative analysis of single-cell RNA sequencing methods (Mol Cell 2017)	10.1016/j.molcel.2017.01.023	Smart-seq2 most sensitive per cell.
Full-length RNA-seq from single cells using Smart-seq2 (Nat Protoc 2014)	10.1038/nprot.2014.006	The protocol used by Segerstolpe.
Benchmarking single-cell mRNA-seq technologies ... cell types with low RNA content (2022)	10.7171/3fc1f5fe.dbeabb2a	Sensitivity in low-RNA cell types.

Gap: I did not find a dedicated single-nucleus vs single-cell comparison for adipose or islets in this search.

J. Cell type annotation

Paper	DOI	Why read it
Automated methods for cell type annotation on scRNA-seq data (Comput Struct Biotechnol J 2021)	10.1016/j.csbj.2021.01.015	Overview of approaches.
Evaluation of cell type annotation R packages on scRNA-seq data (Genomics Proteomics Bioinformatics 2021)	10.1016/j.gpb.2020.07.004	Benchmark of supervised tools.
Generation of human islet cell type-specific identity genesets (Nat Commun 2022)	10.1038/s41467-022-29588-8	Abstract: meta-analysis of scRNA-seq to define alpha/beta/gamma/delta marker sets. Use for islet annotation.
Transcriptomes of the major human pancreatic cell types (Diabetologia 2011)	10.1007/s00125-011-2283-5	Older bulk reference for alpha, beta, duct, acinar.
Integrated single cell and spatial map of human white adipose tissue (Nat Commun 2023)	10.1038/s41467-023-36983-2	Adipose annotation reference (also in section F).

Skipped: LLM-based annotation benchmark (Brief Bioinform 2025, 10.1093/bib/bbaf622). It exists, but I would not rely on it as a beginner.

K. Compositional / differential abundance (T2D may change cell-type proportions)

Paper	DOI	Takeaway from abstract
scCODA: a Bayesian model for compositional single-cell data analysis (Nat Commun 2021)	10.1038/s41467-021-27150-6	Handles compositionality and low sample size.
Milo: differential abundance testing using k-NN graphs (Nat Biotechnol 2022)	10.1038/s41587-021-01033-z	Neighbourhood-level DA without hard clusters.

Paper	DOI	Takeaway from abstract
Benchmarking differential abundance methods (Genome Biol 2024)	10.1186/s13059-023-03143-0	Compares DA methods.
Robust differential composition and variability analysis, sccomp (bioRxiv)	10.1101/2022.03.04.482758	Alternative composition model.
Best practices for single-cell analysis across modalities (Nat Rev Genet 2023)	10.1038/s41576-023-00586-w	Abstract: summarises benchmarks into best-practice workflows. Good umbrella paper to read alongside section A.

L. Pathways and regulator activity

Paper	DOI	Why read it
decoupleR: ensemble of methods to infer biological activities from omics data (Bioinform Adv 2022)	10.1093/bioadv/vbac016	One framework for pathway/TF activity; works on pseudobulk output.
PROGENy: perturbation-response genes reveal signaling footprints (Nat Commun 2018)	10.1038/s41467-017-02391-6	Pathway activity from footprint genes.
Transfer of regulatory knowledge from human to mouse (BBA Gene Regul Mech 2019)	10.1016/j.bbagr.2019.194431	DoRothEA and PROGENy benchmark.

Gap: search for single-cell-specific GSEA tools returned niche packages only. For T2D, running GSEA/decoupleR on pseudobulk DE results (per section D) is the safer route; I have no benchmark paper to cite for that choice.

M. Cell-cell communication

Paper	DOI	Takeaway from abstract
Inference and analysis of cell-cell communication using CellChat (Nat Commun 2021)	10.1038/s41467-021-21246-9	The CellChat method.
CellChat for systematic analysis of cell-cell communication ... (Nat Protoc 2024)	10.1038/s41596-024-01045-4	Protocol, good for a beginner.
Comparison of resources and methods to infer cell-cell communication from single-cell RNA data (Nat Commun 2022)	10.1038/s41467-022-30755-0	Compares resources and methods (LIANA). Read before choosing a tool.
ESICCC: evaluation, selection and integration of CCC inference methods (Genome Res 2023)	10.1101/gr.278001.123	Abstract: benchmarks 18 ligand-receptor methods.
Comparative analysis of cell-cell communication at single-cell resolution (Nat Biotechnol 2023)	10.1038/s41587-023-01782-z	Condition-comparison angle, useful for control vs T2D.

N. Trajectory / dedifferentiation

Paper	DOI	Why read it
A comparison of single-cell trajectory inference methods (Nat Biotechnol 2019)	10.1038/s41587-019-0071-9	Abstract: compares 29 methods.
Cell-state trajectories in T2D beta cells	see Avrahami 2020 and the SMOC1 alpha-cell paper in literature_reading_list.md section E	Applied examples of dedifferentiation analysis.

Index quirk: one search returned a record with this benchmark's abstract attached to an unrelated BMJ title (doi 10.1136/bmj.325.7378.1449). Ignore it; it is a metadata error in the index. Trajectory methods are optional for a first Control vs T2D analysis; treat as advanced.

O. Genetics link (T2D GWAS to cell types)

Paper	DOI	Why read it
Single cell chromatin accessibility reveals pancreatic islet cell type- and state-specific regulatory programs of diabetes risk (Nat Genet 2021)	10.1038/s41588-021-00823-0	Abstract: T2D GWAS enrichment in beta-cell states.
3D chromatin maps of the human pancreas reveal lineage-specific regulatory architecture of T2D risk (Cell Metab 2022)	10.1016/j.cmet.2022.08.014	Variant-to-gene links.
Cell-type-specific cis-eQTLs in pancreatic cell types identify novel risk genes for T2D (Brief Bioinform 2025)	10.1093/bib/bbaf531	Abstract: 328 cell-type-specific cis-eQTLs.
EPIC: inferring relevant cell types for complex traits from GWAS and scRNA-seq (PLoS Genet 2022)	10.1371/journal.pgen.1010251	Method to link GWAS to cell types.

Paper	DOI	Why read it
Implicating T2D effector genes with promoter-focused Capture-C (Diabetologia 2024)	10.1007/s00125-024-06261-x	Metabolic cell models.

Still open

- Single-nucleus vs single-cell comparisons specific to adipose or islets.
- A benchmark specifically supporting pathway analysis choice on single-cell/pseudobulk data.
- Your original items 3 to 5 (message was cut off before them).

Mentor plan, part 1: datasets, papers, 12-week roadmap, first analysis

Status 2026-10-05. Verification level is marked on every claim: VERIFIED = read in paper text; ABSTRACT = abstract only; UNVERIFIED = memory, must be checked. GEO/NCBI/EBI were unreachable from my sandbox, so no GEO page has been read directly.

1. Five datasets for Control vs T2D (human islets)

Rank	Dataset	Donors (ND / T2D)	Cells	Tech	Access	Best for
1	Elgamal 2023, HPAP (doi:10.2337/db23-0130)	65 in final map; 29 ND, 17 T2D of 67 downloaded (also 10 T1D, 9 ND Aab+)	192,203	HPAP scRNA-seq (chemistry not confirmed)	Raw FASTQ via PANC-DB (data-access terms apply). Authors' processed object: UNVERIFIED	Main project
2	Lawlor 2017 (doi:10.1101/gr.212720.116)	8: 5 / 3	1,050 (622 ND, 428 T2D)	Fluidigm C1	SRA SRP075970; processed GEO GSE86473	Learning
3	Seegerstolpe 2016 (doi:10.1016/j.cmet.2016.08.020)	10: 6 healthy / 4 T2D	2,209	Smart-seq2	Portal sandberg.cmb.ki.se/pancreas; E-MTAB-5061 UNVERIFIED	Validation, learning
4	Avrahami 2020 (doi:10.1016/j.molmet.2020.101057)	22 total: 4 adult ND, 10 adult T2D, 8 younger ND	not retrieved	Fluidigm C1	GEO GSE154126	Validation of beta-cell dedifferentiation
5	Bandesh 2025/2026 (doi:10.1101/2025.01.17.633590; 10.1038/s44318-026-00744-w)	not retrieved	not retrieved	not retrieved	not retrieved	Advanced; read methods first

Why not the others:

- Xin 2016 (1,492 cells, GSE81608 UNVERIFIED): donor counts not retrieved; small and older. Fine as extra validation once checked.
- No human adipose dataset with T2D vs control labels is verified. Adipose leads in literature_reading_list.md section F are obesity/insulin-resistance studies, not T2D case-control. Do not use adipose as the main tissue until a labelled dataset is confirmed.
- Donor reality check: Lawlor (3 T2D) and Seegerstolpe (4 T2D) are too small for donor-level inference. Use them to learn the pipeline and for exploratory replication, not as evidence.
- Cell counts per donor are heavily uneven in these plate-based sets; check this before any proportion analysis.

2. Ten papers to read first (in order)

#	Paper	DOI	Read	Learn
1	Current best practices in scRNA-seq analysis: a tutorial (Mol Syst Biol 2019)	10.15252/msb.20188746	Whole	The full pipeline and its decisions.
2	Best practices for single-cell analysis across modalities (Nat Rev Genet 2023)	10.1038/s41576-023-00586-w	Skim, then reference	Benchmark-backed defaults.
3	Confronting false discoveries in single-cell differential expression (Nat Commun 2021)	10.1038/s41467-021-25960-2	Whole	Why cell-level tests fail; pseudobulk.
4	Benchmarking methods for differential states from multi-subject scRNA-seq (Brief Bioinform 2022)	10.1093/bib/bbac286	Whole	Abstract: pseudobulk best overall.
5	Design and power analysis for multi-sample single cell genomics (Nat Commun 2021)	10.1038/s41467-021-26779-7	Methods and figures	Donors, not cells, drive power.
6	Elgamal 2023 integrated islet map (Diabetes)	10.2337/db23-0130	Whole; copy the pipeline	A donor-aware T2D workflow.
7	Lawlor 2017 (Genome Res)	10.1101/gr.212720.116	Results and methods	Your first dataset.

#	Paper	DOI	Read	Learn
8	Segerstolpe 2016 (Cell Metab)	10.1016/j.cmet.2016.08.020	Results	Second dataset; compare findings.
9	Benchmarking atlas-level data integration (Nat Methods 2022)	10.1038/s41592-021-01336-8	Abstract, figures	When integration is needed. Read with the Genome Res 2025 critique (10.1101/gr.279886.124).
10	Milo (Nat Biotechnol 2022) and scCODA (Nat Commun 2021)	10.1038/s41587-021-01033-z ; 10.1038/s41467-021-27150-6	Intro and methods overview	Differential abundance.

Reviews from 2022-2026 on single-cell transcriptomics in diabetes: NOT yet found. I have not searched for reviews specifically; that is the next literature task.

3. Statistical stance (non-negotiable)

10 control + 10 T2D donors with 20,000 cells is $n = 20$, not 20,000. Cells from one donor share genotype, age, BMI, isolation batch, and ambient RNA, so they are nested within donors and correlated.

- Defensible for publication: pseudobulk (sum counts per donor per cell type) with edgeR, DESeq2 or limma-voom, with donor as the unit; or mixed models with a donor random effect, where the literature (see reading list section D) argues they perform comparably if specified well. Include covariates (sex, age, BMI, batch) when donor numbers allow.
- Not defensible: Wilcoxon or t-test across cells, reporting cell-level p-values as evidence of disease effect. Fine for marker finding within one dataset, not for Control vs T2D.
- Permutation tests: permute donor labels, never cell labels.
- Abundance: use donor-level proportions with a compositional method (scCODA, Milo, sccomp); proportions sum to 1, so one cell type rising makes others fall.

4. 12-week roadmap

Wk	Concepts	Read	Do	Output
1	DNA/RNA, transcription, exons/introns, RNA-seq, UMI, count matrix, MT/ribosomal genes	Paper 1 sections on experimental setup	Open a count matrix in pandas, compute library size, genes per cell	One-page glossary in your own words
2	Islet biology: alpha/beta/delta, insulin, glucose handling, T2D: beta-cell dysfunction, ER and oxidative stress, inflammation	Lawlor intro/discussion; Avrahami abstract	List marker genes per islet cell type (INS, GCG, SST, PPY, PRSS1, KRT19)	Marker table with sources
3	Cell vs nucleus, Smart-seq2 vs 10x, dropout, sparsity, depth, batch and donor effects	Paper 1; technology comparison papers (reading list part 2, section I)	Load Lawlor into AnnData; explore sparsity	Plots: counts/cell, genes/cell, MT fraction
4	QC, normalization, HVGs	Paper 1 QC/normalization sections	Run QC filters; justify thresholds per donor	Before/after QC figures, written thresholds
5	PCA, kNN graph, Leiden resolution, UMAP	Paper 1 clustering section	Cluster at several resolutions; check stability	Clustering notebook; resolution sweep
6	Marker genes, annotation	Islet identity genesets paper (10.1038/s41467-022-29588-8)	Annotate cell types; check for doublets and ambient hormone signal	Annotated UMAP, dotplot
7	Pseudoreplication, pseudobulk	Papers 3, 4	Pseudobulk by donor x cell type; edgeR/limma or pydeseq2; compare with a naive cell-level test	Volcano plots, inflation comparison
8	Power, abundance	Papers 5, 10	Donor-level proportions; scCODA or Milo	Abundance plot with donor points
9	Multiple testing, GSEA/GO/Reactome, decoupleR	decoupleR (10.1093/bioadv/vbac016)	Pathways on pseudobulk results	Enrichment figure, table
10	Replication	Segerstolpe paper	Repeat steps on Segerstolpe; compare direction of effects	Concordance plot, short report
11	Batch/integration, HPAP structure	Paper 9 and Genome Res critique	Request PANC-DB access; read Elgamal methods	Access request, data-plan note
12	Project design	Paper 6	Write aims and analysis plan	3-page proposal for your PI

5. Recommended first dataset and analysis

Dataset: Lawlor 2017 (GSE86473 processed; SRA SRP075970 raw). Reason: small, both labelled groups, plate-based so no droplet ambient-RNA complication, and the accession was read in the paper itself. Weakness: only 3 T2D donors.

First analysis:

1. Load the processed matrix; confirm the donor and diagnosis labels from GEO sample metadata (I could not read them).
2. Per-donor QC, then normalize, HVGs, PCA, kNN, Leiden, UMAP.
3. Annotate alpha, beta, delta, PP, acinar, ductal; color UMAP by donor and diagnosis to see donor effects.
4. Pseudobulk by donor x cell type; test beta cells T2D vs ND with 5 vs 3 donors. Treat as exploratory.
5. Also run the naive cell-level test and compare; the gap is the lesson on pseudoreplication.

Tool choice: start with Scanpy/AnnData (you already know Python), and learn R only for the pseudobulk step if needed (edgeR/limma-voom or muscat are the field standard; I believe pydeseq2 is a Python alternative but I have not verified it this session). Learn Seurat later, mainly to read other people's code.

Open items

- Verify GEO metadata for GSE86473, GSE154126 and E-MTAB-5061, and the HPAP access path.
- Search for 2022-2026 reviews on single-cell transcriptomics in diabetes.
- Find a T2D-labelled adipose or other metabolic-tissue dataset.
- Remaining deliverables: top-10 dataset list, full workflow, aims, portfolio structure, mistakes list, 50-term glossary, step-by-step teaching notes.

Mentor plan, part 2: remaining deliverables

Verification labels: VERIFIED = read in paper text; ABSTRACT = abstract only; UNVERIFIED = memory, check before use. Deliverable 3 (12-week roadmap) is in mentor_plan_part1.md. Links are in links.md.

Deliverable 1: ranked public datasets for Control vs T2D

I cannot honestly give 10 verified datasets. I verified 4 from paper text and have leads for the rest. Do not treat leads as confirmed.

Rank	Dataset	Tissue	Donors (ND/T2D)	Cells	Status	Role
1	HPAP, Elgamal 2023, doi:10.2337/db23-0130	Islets	29 / 17 of 67 downloaded (65 in final map)	192,203	VERIFIED	Main
2	Lawlor 2017, GSE86473	Islets	5 / 3	1,050	VERIFIED	Learning
3	Segerstolpe 2016	Islets	6 / 4	2,209	VERIFIED counts; accession UNVERIFIED	Learning, replication
4	Avrahami 2020, GSE154126	Islets	4 adult ND + 4 younger groups / 10 T2D	not retrieved	VERIFIED donors	Replication
5	Bandesh 2025/2026, doi:10.1038/s44318-026-00744-w	Islets	not retrieved	not retrieved	ABSTRACT	Advanced
6	Xin 2016, GSE81608	Islets	not retrieved	1,492	ABSTRACT; accession UNVERIFIED	Extra replication
7	Camunas-Soler 2020 patch-seq, doi:10.1016/j.cmet.2020.04.005	Islets, with electrophysiology	not retrieved	not retrieved	ABSTRACT	Advanced (links function to expression)
8	Alpha-cell scRNA+snRNA-seq, doi:10.1038/s41467-025-62670-5	Islets	not retrieved	not retrieved	ABSTRACT	Advanced; snRNA-seq example
9	Sex differences, 52 donors with and without T2D, doi:10.21203/rs.3.rs-4607352/v1	Islets, scRNA + snATAC	52 total	not retrieved	ABSTRACT (preprint)	Validation if data are public
10	Adipose: no verified T2D-labelled set	Adipose	none	none	Gap	Leads in literature_reading_list.md F

Unsuitable or caution:

- Datasets without diabetes labels (e.g. healthy-only islet atlases, mouse atlases) cannot answer your question.
- Mouse db/db or HFD islet data are not human T2D; use only for hypothesis support.
- Obesity-only adipose snRNA-seq studies do not isolate T2D; any adipose finding would be confounded by BMI.

Deliverable 2: 18 papers and resources (priority order)

Full DOIs and PMC links are in links.md. Short list: 1 Luecken & Theis best practices (whole). 2 Best practices across modalities (skim). 3 Confronting false discoveries (whole). 4 Multi-subject DS benchmark (whole). 5 Multi-sample power/design (methods, figures). 6 Elgamal 2023 (whole, copy the pipeline). 7 Lawlor 2017 (results, methods). 8 Segerstolpe 2016 (results). 9 Atlas-level integration benchmark + Genome Res 2025 critique (abstract, figures). 10 Milo + scCODA (methods overview). 11 Avrahami 2020 (dedifferentiation result). 12 Islet identity genesets (annotation markers). 13 decoupleR (tutorial). 14 CellChat Nat Protoc 2024 (protocol; optional). 15 Comparison of CCC methods (Nat Commun 2022; optional). 16 Bandesh 2025/2026 (methods, once you are on HPAP). 17 Camunas-Soler patch-seq (advanced). 18 Human adipose integrated map (adipose context). Reviews from 2022-2026 on single-cell transcriptomics in diabetes: NOT yet found; search PubMed for "single-cell diabetes review" before citing any.

Deliverable 4: end-to-end workflow

Teaching format for each step: what, why, assumptions, failure modes, validation, expected plot, how to say it to your PI.

4.1 Data and metadata

- What: one donor table (ID, diagnosis, sex, age, BMI, batch, isolation site, culture time) and one count matrix per dataset.
- Why: all inference is donor-level; metadata drives the model.
- Fail: batch perfectly aligned with diagnosis (then disease and batch cannot be separated).
- Validate: cross-tab diagnosis x batch x sex.
- Plot: donor-level bars of cell counts per donor.
- PI line: "Here are the donors; this is what we can and cannot adjust for."

4.2 QC

- What: remove empty droplets, dying cells, doublets; correct ambient RNA in droplet data.
- Why: low-quality cells form fake clusters; ambient INS/GCG makes alpha cells look INS-positive and creates false DEGs (ambient-RNA abstracts in `literature_reading_list.md` section B).
- Assumptions: thresholds are tissue-specific; do not copy PBMC cutoffs. Check MT fraction per donor and cell type.
- Fail: filtering with one global threshold that removes a whole donor or cell type.
- Validate: per-donor distributions before/after; cells removed per donor.
- Plot: violin of genes/UMIs/MT% per donor.
- PI line: "QC removed X% of cells, evenly across diagnosis."

4.3 Normalization

- What: put cells on a comparable scale. Standard: divide by total counts, scale, log1p. Alternatives: Pearson residuals; for Smart-seq2, length-normalized values.
- Why: cells differ in capture efficiency and depth, not biology.
- Assumption: most genes are unchanged and cell types have similar total RNA. Violated when a cell type has much more RNA.
- Fail: normalized values used for pseudobulk testing (use raw counts there).
- Validate: library size no longer drives PC1.
- PI line: "Normalization removes depth, not biology."

4.4 HVG, PCA, neighbours, clustering, UMAP

- HVG: keep genes with variance beyond expectation (about 2000), reducing noise.
- PCA: compress to 30-50 components; check an elbow plot and whether PCs track donor rather than biology.
- kNN graph: each cell links to its k nearest neighbours in PC space (try k 15-30).
- Leiden: finds densely connected groups on that graph. Resolution sets granularity: higher means more, smaller clusters.
- UMAP: 2D picture for display only. Distances and cluster sizes in UMAP are not quantitative; never test on UMAP coordinates.
- Fail: choosing resolution until you see a "T2D cluster" (circular).
- Validate: run several resolutions; check cluster stability, marker specificity, and that clusters contain multiple donors.
- Plot: UMAP coloured by cell type, donor, diagnosis side by side.
- PI line: "Clusters are hypotheses until markers and replication support them."

4.5 Annotation

- Use canonical markers (INS beta, GCG alpha, SST delta, PPY PP, PRSS1 acinar, KRT19 ductal, plus endothelial/stellate/immune) and the islet identity genesets paper; optionally reference mapping.
- Validate: dotplot; hormone ambient check; compare proportions with the original paper.

4.6 Cell-type abundance

- Unit: donor. Compute proportions per donor, then use scCODA, Milo (donor as replicate) or sccomp.

- Why: proportions sum to 1, so one type rising makes others fall.
- Fail: chi-square test on cell counts (pseudoreplication).
- Plot: per-donor proportions as points with group means.

4.7 Differential expression: Control vs T2D

- Default: pseudobulk. For each cell type, sum raw counts per donor; filter cell types with too few cells per donor and too few donors per group; model counts with edgeR (quasi-likelihood), DESeq2 or limma-voom; design: diagnosis + sex + age (+ batch/BMI if donors allow).
- Alternative: mixed model with donor random effect (e.g. NEBULA, glmmTMB, dreamlet). The Bioinformatics 2024 paper abstract says pseudobulk with proper offsets matches GLMMs; use mixed models mainly when donors contribute varying cell numbers and you want covariates handled per cell.
- Report: log2 fold change, raw p, FDR (Benjamini-Hochberg) per cell type, and decide how to handle multiplicity across cell types.
- Fail: p-values from per-cell Wilcoxon; treating UMAP separation as evidence; unadjusted covariates.
- Validate: p-value histogram is flat plus a spike near 0; MA/volcano plots; leave-one-donor-out stability; permutation of donor labels.
- Elgamal found 84 T2D genes across cell types with 17 T2D vs 29 ND (VERIFIED), so expect modest effects, not hundreds.
- PI line: "Effective sample size is donors; here is the donor-level result."

4.8 Pathways and regulators

- Run GSEA/GO/Reactome/KEGG on the ranked pseudobulk statistics (not on a thresholded list when possible); decoupleR for TF and pathway activity.
- Fail: interpreting enrichment from a ranked list built on cell-level p-values.

4.9 Replication

- Repeat the identical pipeline in an independent cohort; report direction concordance and effect-size correlation of DEGs, not just overlap counts.
- Caveat: platforms differ (Smart-seq2/C1 vs droplet), so compare signs and ranks, not magnitudes. Small cohorts have little power, so lack of replication is weak evidence.

4.10 Advanced, later

- Trajectory/pseudotime, RNA velocity, integration across datasets, scATAC, spatial. Not needed for the first project.

Deliverable 5: research question and aims

Question: How does type 2 diabetes alter cell-type composition and cell-type-specific transcription in human pancreatic islets, and are these changes reproducible across independent donor cohorts?

- Aim 1, cell-type identification: annotate islet cell types in HPAP using canonical markers; evaluate against published labels. Analysis: clustering stability, marker specificity.
- Aim 2, abundance: test donor-level proportion differences (scCODA/Milo/sccomp), adjusting for sex and age. Statistics: compositional model with donor as replicate; FDR across cell types.
- Aim 3, cell-specific DE: pseudobulk edgeR/limma-voom per cell type with covariates. Statistics: FDR per cell type; leave-one-donor-out sensitivity.
- Aim 4, pathways: GSEA/decoupleR on ranked pseudobulk statistics for beta, alpha, delta cells; check stress, inflammation, mitochondrial themes against literature.
- Aim 5, replication: repeat Aims 3-4 in Lawlor, Segerstolpe and Avrahami; test directional concordance (sign test, rank correlation).

Realism check: HPAP access and FASTQ processing may take weeks. A safe internship scope is Aims 1-3 on one dataset plus Aim 5 on a small one.

Deliverable 6: portfolio structure

t2d-islet-sc/	
README.md	question, data, key results, how to run
environment.yml	pinned versions (python, scanpy, edgeR via rpy2 or R env)

```

data/
  README.md          source URL, accession, download date, checksums
  raw/               never edited (git-ignored)
  metadata/donors.csv donor table
notebooks/
  01_qc.ipynb ... 08_replication.ipynb
src/                 reusable functions (QC, pseudobulk)
results/figures, results/tables
reports/             2-page summary, slide deck for PI
docs/                literature notes, decisions log (why each threshold)

```

Practices: fixed random seeds, one notebook per step, a decisions log, no large data in git, a short methods paragraph per analysis.

Deliverable 7: common mistakes (statistically invalid ones first)

Invalid:

1. Treating cells as independent replicates (pseudoreplication).
2. Using per-cell Wilcoxon p-values to claim Control vs T2D differences.
3. Permuting cell labels instead of donor labels.
4. Chi-square on cell counts for abundance.
5. Pseudobulk built from normalized/logged values instead of raw counts.
6. Batch or site perfectly confounded with diagnosis, and not stating it.
7. Clustering, then testing the clusters for diagnosis differences on the same data without acknowledging circularity (Cell Syst 2019 paper on post-clustering inference).
8. Integration that removes the disease signal, then claiming no effect.
9. Ignoring multiple testing across genes and cell types.
10. Reporting a cell type with 3 cells in a donor as a donor-level measurement.

Common technical:

11. Copying QC thresholds from other tissues.
12. Ignoring ambient hormone RNA in islets.
13. Reading UMAP distances and cluster sizes quantitatively.
14. Tuning resolution to get the cluster you want.
15. Annotating by one marker gene.
16. Ignoring sex, age, BMI and donor-level covariates.
17. Overinterpreting a p-value without effect size or replication.
18. Skipping donor-level plots.
19. Reporting pathway enrichment from an arbitrary thresholded list.
20. Mixing Smart-seq2 and droplet data without addressing technology effects.

Deliverable 8: glossary (50 terms)

1 Transcriptome: all RNA in a sample. 2 Gene expression: RNA produced from a gene. 3 Transcript: an RNA copy of a gene. 4 Exon: retained coding segment. 5 Intron: removed segment (reads from nuclei often include introns). 6 mRNA. 7 Poly(A): tail used for capture in most protocols. 8 Read: one sequenced fragment. 9 FASTQ: raw read file with quality. 10 Alignment: mapping reads to a genome. 11 UMI: tag that counts original molecules. 12 Cell barcode: tag identifying the cell. 13 Count matrix: genes x cells table of counts. 14 Library size: total counts in a cell. 15 Sequencing depth: reads per cell. 16 Dropout: gene present but not detected. 17 Sparsity: fraction of zeros. 18 scRNA-seq: whole cells. 19 snRNA-seq: nuclei; used for tissues like adipose. 20 Smart-seq2: full-length plate protocol. 21 10x Chromium: droplet protocol. 22 Droplet: oil bubble holding one cell. 23 Empty droplet: no cell, ambient RNA only. 24 Doublet: two cells in one barcode. 25 Ambient RNA: free RNA contaminating droplets. 26 QC metrics: genes, UMIs, MT fraction. 27 Mitochondrial genes: MT- genes; high fraction suggests damage. 28 Ribosomal genes. 29 Normalization. 30 Log1p. 31 HVG: highly variable genes. 32 PCA. 33 kNN graph. 34 Leiden/Louvain: community detection. 35 Resolution. 36 UMAP/t-SNE: 2D visualization. 37 Marker gene. 38 Cell type annotation. 39 Batch effect: technical difference between runs. 40 Donor effect: differences between individuals. 41 Biological replicate: independent donor/animal. 42 Pseudoreplication. 43 Pseudobulk: summed counts per donor per cell type. 44 Differential expression. 45 Log2 fold change. 46 p-value. 47 FDR / adjusted p-value (Benjamini-Hochberg).

48 Mixed-effects model: includes donor random effect. 49 Differential abundance / compositional data. 50 GSEA/gene set enrichment, with TF activity and cell-cell communication as the next-layer interpretations. Extra for T2D biology (not counted): beta cell, alpha cell, delta cell, PP cell, insulin, glucagon, somatostatin, insulin resistance, beta-cell dedifferentiation, ER stress, oxidative stress, islet, HbA1c.

Open items

- Verify GEO metadata for GSE86473, GSE154126; E-MTAB-5061; HPAP access path.
- Find 2022-2026 diabetes single-cell reviews.
- Find a T2D-labelled adipose or other metabolic dataset.
- Confirm pydeseq2/NEBULA/dreamlet specifics from official documentation before using.